

Attempt 1



In Progress

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3 Attempts Allowed

Details

Overview

Finalize your predictive modeling project by integrating the feedback and findings from Milestone 1 (Analytic Plan) and Milestone 2 (Preliminary Results). Present a comprehensive, client-ready deliverable demonstrating how you refined data preparation, advanced your modeling approach, derived feature importance insights, and developed business-oriented recommendations.

Instructions

1. Consolidate Data Preparation & Feature Engineering Updates

- Summarize any refinements made post-Milestone 2 (e.g., outlier removal, additional feature creation).
- Discuss how these changes improved model performance or interpretability.

2. Expand Modeling & Evaluation

- Implement at least one advanced method (e.g., Random Forest, Gradient Boosting) in addition to the models from Milestone 2.
- Tune hyperparameters (e.g., grid search, randomized search) to optimize each model's performance.
- Compare all models (including those from Milestone 2) using detailed metrics such as AUC, precision/recall, F1-score, and visualizations like ROC or precision-recall curves.

3. Assess Feature Importance

- Use an interpretability method (e.g., SHAP values, model-based feature importance, or coefficient analysis) for the final model.
- Explain which features are most critical for the churn (or other business) outcome and why these insights matter.

4. Offer Business Recommendations

- Propose actionable strategies (e.g., retention campaigns, personalized outreach) based on high-risk customer segments.
- Discuss deployment feasibility (batch vs. real-time) and the organizational steps needed for implementation.
- Note any limitations or future improvements (e.g., additional data sources, more advanced modeling) to further refine your solution.

5. Compile Appendices

- Provide **supporting materials** such as hyperparameter tuning results, extended tables, and relevant code snippets.
- Keep your **main report concise**, focusing on the most critical findings and insights.

Deliverable

1. Restated Data Preparation and Feature Engineering

- Summarize any **post-Milestone 2** refinements (e.g., removing outliers, creating new features).
- Highlight how updates **enhanced** performance or interpretability.

2. Modeling and Evaluation

- Incorporate an advanced method (e.g., Random Forest, Gradient Boosting) along with earlier models.
- Tune hyperparameters and compare all models using metrics like AUC, precision/recall, and F1-score.
- Include visualizations (ROC curves, precision-recall plots) to illustrate performance.

3. Feature Importance Analysis

- Use SHAP values, model-based importance, or coefficient analysis to identify key predictors.
- Discuss business implications of these critical features.

4. Business Recommendations

- Provide actionable strategies for using the final model (e.g., targeted retention).
- Consider deployment readiness and outline next steps for long-term model maintenance.
- Acknowledge limitations and potential avenues for improvement.

5. Appendices

- Present **supporting materials** (hyperparameter search details, extended tables, code snippets).
- Only include items that **support** the findings in the main body.

Formatting and Submission

- **Length:** No more than 20 single-spaced pages (excluding appendices).
- **Format:** MS Word/PDF (e.g., from RMarkdown); 12-point Times New Roman font.
- **Submission:**
 1. Upload your file to Canvas (either by selecting **Upload** or dragging and dropping the file).
 2. Click **Submit Assignment** to finalize.

Important:

- Include only the charts/graphs and analytic outputs that support your findings and discussion.
- Omitting necessary evidence or including irrelevant outputs will negatively impact the grade.
- Be thoughtful and purposeful—your final report should clearly demonstrate how the modeling insights meet the business need and inform actionable decision-making.

Class Project Milestone 3 Rubric			
Criteria	Ratings		Points
Restated Data Preparation & Feature Engineering view longer description	Full Marks 20 pts	No Marks 0 pts	/20 pts
Modeling & Evaluation view longer description	Full Marks 40 pts	No Marks 0 pts	/40 pts
Feature Importance Analysis view longer description	Full Marks 20 pts	No Marks 0 pts	/20 pts
Business Recommendations view longer description	Full Marks 15 pts	No Marks 0 pts	/15 pts
Appendices & Overall Report Quality view longer description	Full Marks 5 pts	No Marks 0 pts	/5 pts

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